

PENETRATION TEST REPORT

Guided Pentest: Infrastructure

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Table of Contents

- 1. Executive Summary
- 2. Technical Summary
- 3. Scope
- 4. Methodology
- 5. Vulnerability Summary
- 6. Detailed Findings
- 7. Conclusion

1. Executive Summary

A penetration test was conducted against a newly deployed infrastructure device belonging to the client, TryHatMe. During the assessment, multiple critical vulnerabilities were identified that resulted in complete compromise of the target system.

An outdated and vulnerable version of UnrealIRCd was exposed externally, allowing remote attackers to gain unauthorized shell access. Following initial compromise, sensitive credentials were discovered stored insecurely in plaintext, enabling escalation to full administrative access.

Immediate remediation is strongly recommended.

2. Technical Summary

The assessment identified UnrealIRCd 3.2.8.1 running on TCP port 6667. This version contains a known backdoor vulnerability allowing unauthenticated remote command execution.

Using a public Metasploit module, shell access was obtained as the 'webmaster' user. Post-exploitation enumeration revealed a plaintext credential file containing the root password, enabling full system compromise.

3. Scope

Target	Description
10.49.179.83	Linux Infrastructure Device

4. Methodology

- Reconnaissance and Enumeration
- Service Identification
- Vulnerability Analysis
- Exploitation

- Post-Exploitation Enumeration
- Privilege Escalation
- Reporting

5. Vulnerability Summary

Vulnerability	Severity	Impact
UnrealIRCd 3.2.8.1 Backdoor Remote Code Execution	Critical	Remote system compromise
Root Password Stored in Plaintext	Critical	Full root compromise

6. Detailed Findings

Finding 1 — UnrealIRCd Backdoor Remote Code Execution

- Severity: Critical

The target exposed UnrealIRCd version 3.2.8.1 on TCP port 6667. This version contains a publicly known backdoor vulnerability allowing unauthenticated remote command execution.

- Impact:
 - Remote command execution
 - Unauthorized shell access
 - Potential lateral movement
 - Privilege escalation opportunities
- Exploitation Steps:
 1. Enumerate services using Nmap.
 2. Identify UnrealIRCd 3.2.8.1.
 3. Research exploits using Searchsploit.
 4. Use Metasploit exploit module.
 5. Obtain reverse shell access.

Proof of Exploitation:

uid=1001(webmaster) gid=1001(webmaster)

- Recommendations:
 - Upgrade UnrealIRCd immediately.
 - Remove outdated software.
 - Restrict unnecessary exposed services.
 - Implement patch management.

Finding 2 — Root Password Stored in Plaintext

- Severity: Critical

The root password was stored in plaintext within /etc/password.txt. The file was readable by low-privileged users e.g webmaster, enabling privilege escalation.

- Impact:
 - Exposure of root credentials
 - Full administrative compromise
 - Complete loss of confidentiality, integrity, and availability
- Exploitation Steps:
 1. Obtain low-privileged shell access.
 2. Read /etc/password.txt.
 3. Retrieve root credentials.
 4. Authenticate via SSH as root.

Proof of Exploitation:

uid=0(root) gid=0(root)

- Recommendations:
 - Remove plaintext password files immediately.
 - Rotate exposed credentials.
 - Use secure password storage mechanisms.
 - Restrict sensitive file permissions.
 - Conduct credential exposure audits.

7. Conclusion

The assessment demonstrated that the target system could be fully compromised through a chain of critical vulnerabilities. Immediate remediation should be prioritized to reduce the likelihood of real-world exploitation.